

# C5\_6. Key concepts for studying Social Media



Universitatea  
Transilvania  
din Braşov

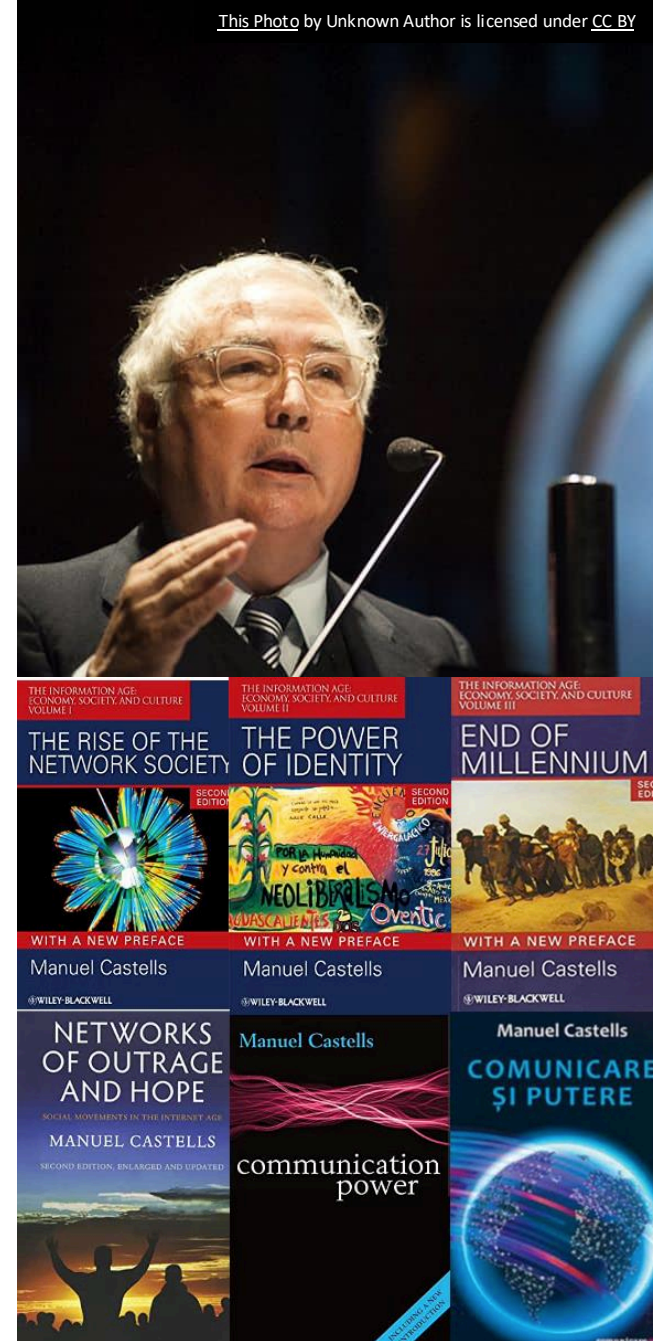
FACULTATEA DE SOCIOLOGIE  
ŞI COMUNICARE



Assoc. Prof. Arabela BRICIU

# Castells (1996,1997, 1998, 2009, 2013)

- The information age - capitalism focuses on information and knowledge
- information processing and communication technologies shape economic, social, and cultural life.
- <<network society>>
- Widespread access to telecommunications technologies has changed the way we live, think and act.
- Networked social relations are a dynamic form of self-developing human activity → transforming all social and economic life spheres.
- Networks have become the predominant form of organisation in all areas of human activity.
- Social power can also be accumulated in cyberspace by marginalised groups in order to challenge state institutions and the global elite.

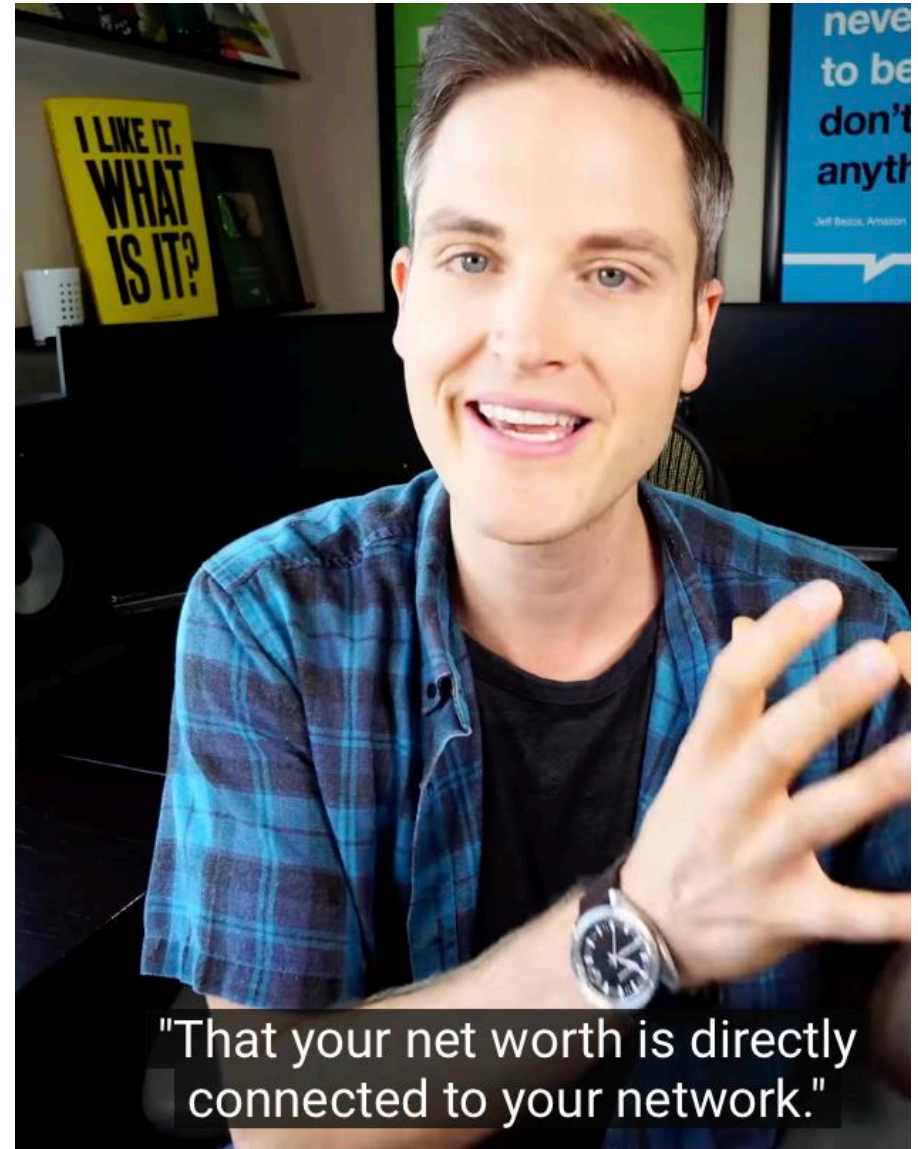




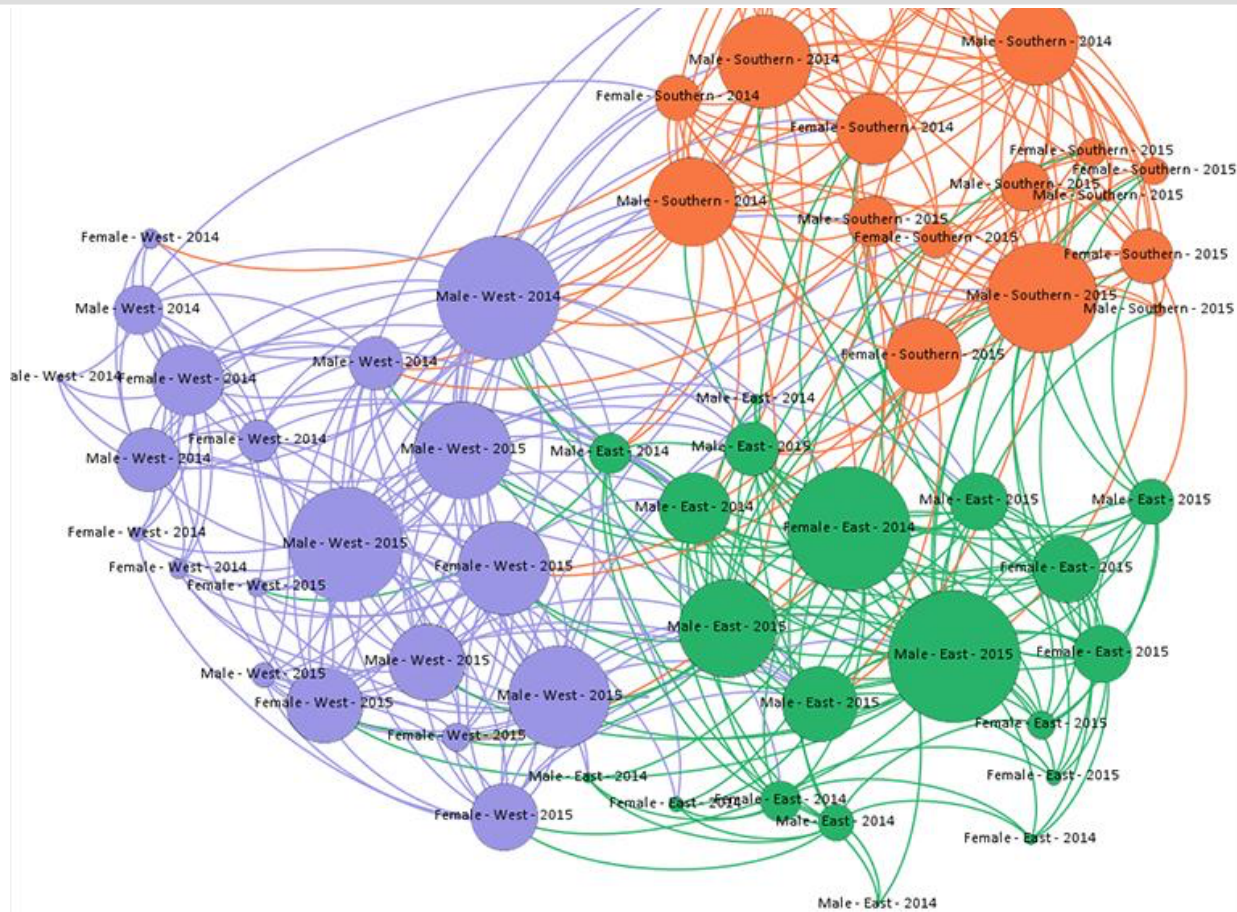
Jordan Peterson - Advice for People Who Aren't Social



An introvert's guide to networking | Rick Turoczy | TEDxPortland



Social Circle — 3 Networking Tips



<https://www.irex.org/insight/visualizing-what-connects-us-social-network-analysis-me>

## Relation - Social ties



- Relation - a series of social ties (of a particular type) between group members.
  - **Trust** - a positive expectation that the other person will behave according to your expectations (**vs. suspicion, doubt**)
  - **Cooperation** - an action with multiple actors and a common goal or set of goals ( vs. **conflict**)
  - **Tolerance** –cohabitation agreement; acceptance and understanding for those who act differently than you, based on other beliefs or lifestyles (vs. **intolerance**)
- → The ties are diverse and can indicate:
  - The way one person relates to another (*e.g.*, friendship, respect, condescension/superiority)
  - The flow of some resources (*e.g.*, circulation of information, goods, money)
  - Affiliation (*e.g.*, membership in an association, group, party; institutional affiliation)
  - Interaction (*e.g.*, exchange of messages, dialogue, face-to-face meeting, fight)

# Social networks

- Social network - a finite set of actors together with the established relationship or relationships between these actors (social relationships that have very different content - information, power, control, etc.)

Actors (individual or group entities) - Nodes

Relationships - Ties

## A. Continuous social ties

### A1. Similarities

- Physical proximity (**e.g.**, neighbourhood neighbours, bank colleagues, next-door neighbours)
- Co-membership (**e.g.** members of the same organisation - party, association, etc.)
- Sharing the same values, behaviours, attitudes, beliefs (e.g. support for right-wing political values, support for the LGBT community, embracing the same religion)

### A2. Social Networks

- Role relationships (**e.g.**, mother, father, boss, teacher, tutor, mentor, etc.)
- Cognitive/emotional relationships (e.g., I am friends with X, I like Y, I know Z, I don't like W, etc.)

## B. Discreet social ties

### B1. Interactions

Talk with...; Dine with...; Go to party with...; Fight with... etc.

### B2. Flows

Resources that flow from one network node (individual) to another in succession (e.g. a sum of money, a book, a piece of information) or simultaneously (e.g. an e-mail can reach several nodes simultaneously).

# Types of social networks

- **Dyad:** two actors and their possible relationships [basic unit in analysis].



the characteristics of ties (directional, reciprocity)

multiplex nature (A, B - friends but also co-workers)

- **Triad:** set of three actors and their possible relationships



**Transitivity:** if A likes B and B likes C, then A will like C.

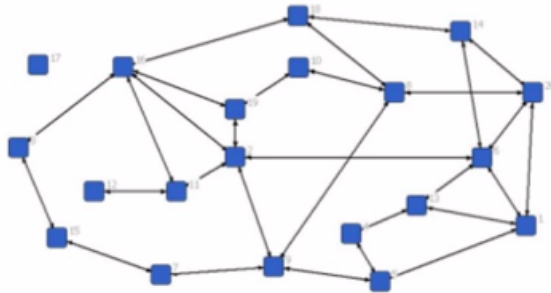
**C** **Balance:** if A and B like each other, then they will be similar in their assessment of C;

if A and B dislike each other, then they will have different assessments of C.

- **One-mode networks** – a single set of actors (the communication network within an organisation)
- **2-mode networks** – two sets of actors : affiliate networks - a set of actors and a set of events
- **Ego-network** – personal networks: consist of a central node (ego) and all nodes to which it is connected (alters)
- **Full-network**

### One-mode

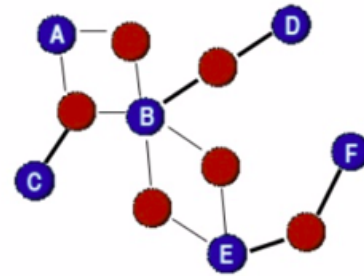
All nodes are the same



Students enrolled in the Social Media course

### 2-mode

Nodes are of two types



Transilvania University of Brasov students  
(from different faculties) participating in  
*Orientation Days*

Football tournaments between company  
departments

## *One-mode and 2-mode networks*

# Terms used in network graphs

- Actors in a social network are often called **NODES** (in the international literature: "Nodes" or "Vertices") (Izquierdo, 2006).
- **RELATIONSHIPS** between nodes are sometimes called "arcs", "links" or "edges", and in the specialized literature: *relationship, link, edge, or tie* (Abbasi et al., 2011).
- The diagrams are named differently
  - GRAPHS
  - SOCIOGRAPHS
  - SOCIAL NETWORK DIAGRAMS, according to Borgatti et al., (2013)

<https://cambridge-intelligence.com/use-cases/social-networks/>

## Graph and timeline visualization to analyze social networks

### The social network visualization challenge

Simply put, a social network is a structure of social actors joined together by connections.

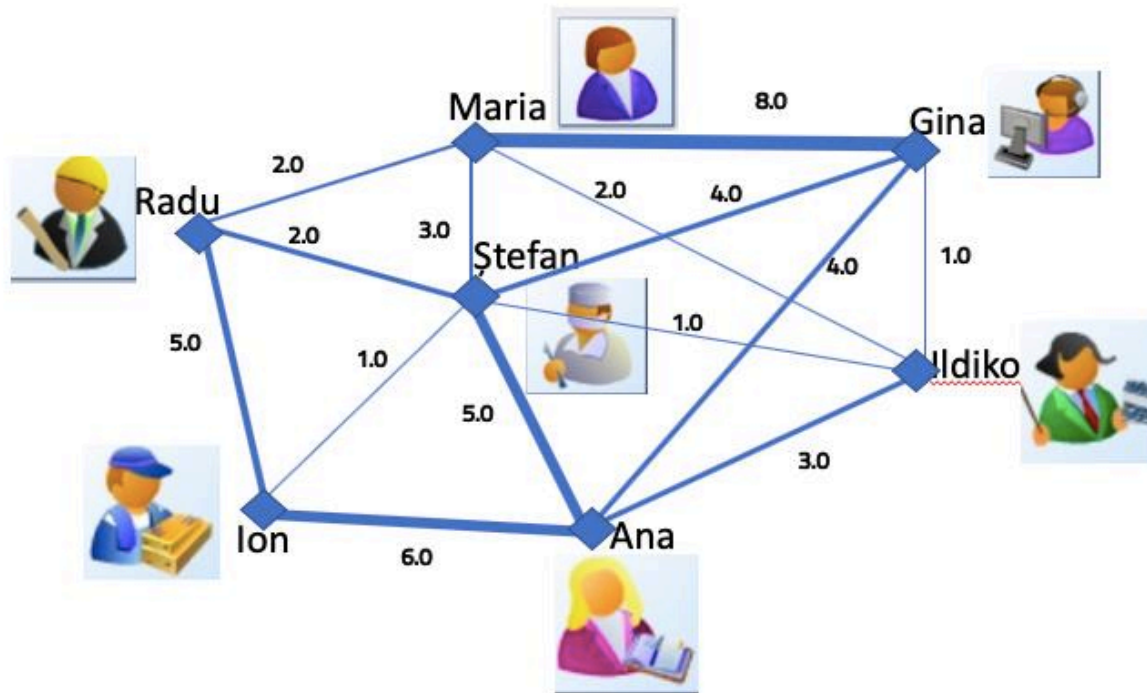
If you understand this structure, you have a much deeper knowledge than if you assessed the social actors in isolation.

Powerful data visualization techniques cut through noisy social network data and reveal parts of the network that need attention.



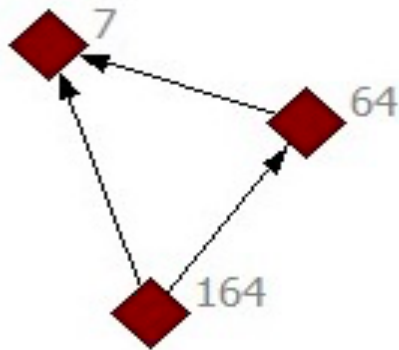
# Social network ties

- Based on **occurrence**: absent or present
- "Distances" and "positions" have specific definitions given by the connections between actors.
- The ties present can be **valorised**, depending on their **intensity**.

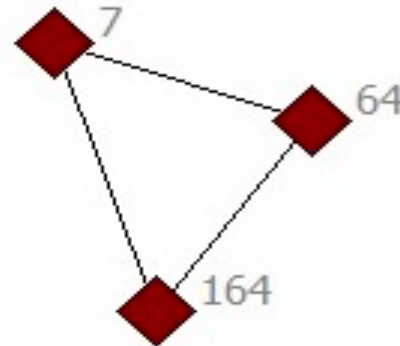


# Social network ties

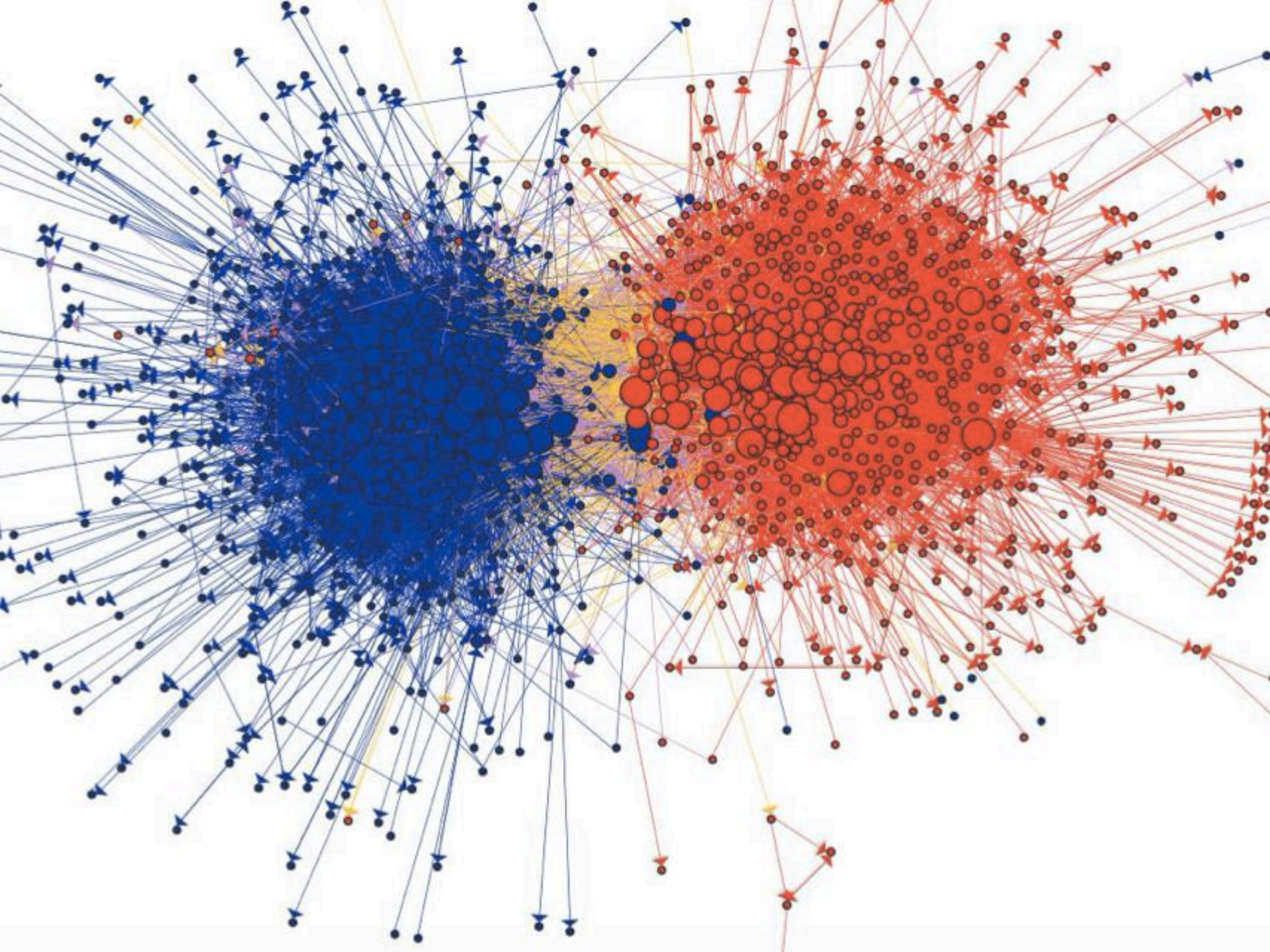
- **Indirect** (e.g. working together)
- **Directed** (e.g. granting approval).



Directed



Indirect



# Social capital



Cooperation



Trust



Social networks  
(ties)



SOCIAL  
CAPITAL

- It is "the aggregate of the actual or potential resources which are linked to possession of a durable network of more or less institutionalized relationships of mutual acquaintance or recognition" (Bourdieu, 1986, p.248)
- Social capital is about "relationships of trust and shared values" (Putman, 2000, p.312)
- „investment in social relations with expected returns" (Lin, 1999, p.30).
- „The main components of social capital are trust, norms of reciprocity, and engagement in networks" (van Schaik 2002, p.7).

<https://youtu.be/T01lQGcvZqY?si=kMBxn9yq2m-35IWw>



SOCIAL CAPITAL  
RESEARCH

**ROBERT  
PUTNAM**



# Capital social

Coleman (1988, 1990): social capital is that type of resource which leads to the constitution of the public good as a result of the voluntary cooperation of individuals, which allows access to resources controlled by others, which reduces the transaction costs by establishing trust and reciprocity between partners, and which provides rapid information exchange.



Social capital is a characteristic of the social structure, which is manifested through trust and relationships that allow individuals, communities and societies to access and mobilize resources and information that would otherwise be removed from them and use them in individual, community and societal development projects (Voicu, 2015, p. 61)



→ Social capital = resources (material, informational, symbolic) that you get from social relations (of different intensities) and that you use for personal (individual) or group purposes - based on trust and reciprocity

<https://youtu.be/bxhqVR-bc3A?si=wHdkdh8SvMuW0J93>

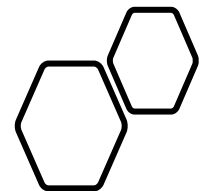
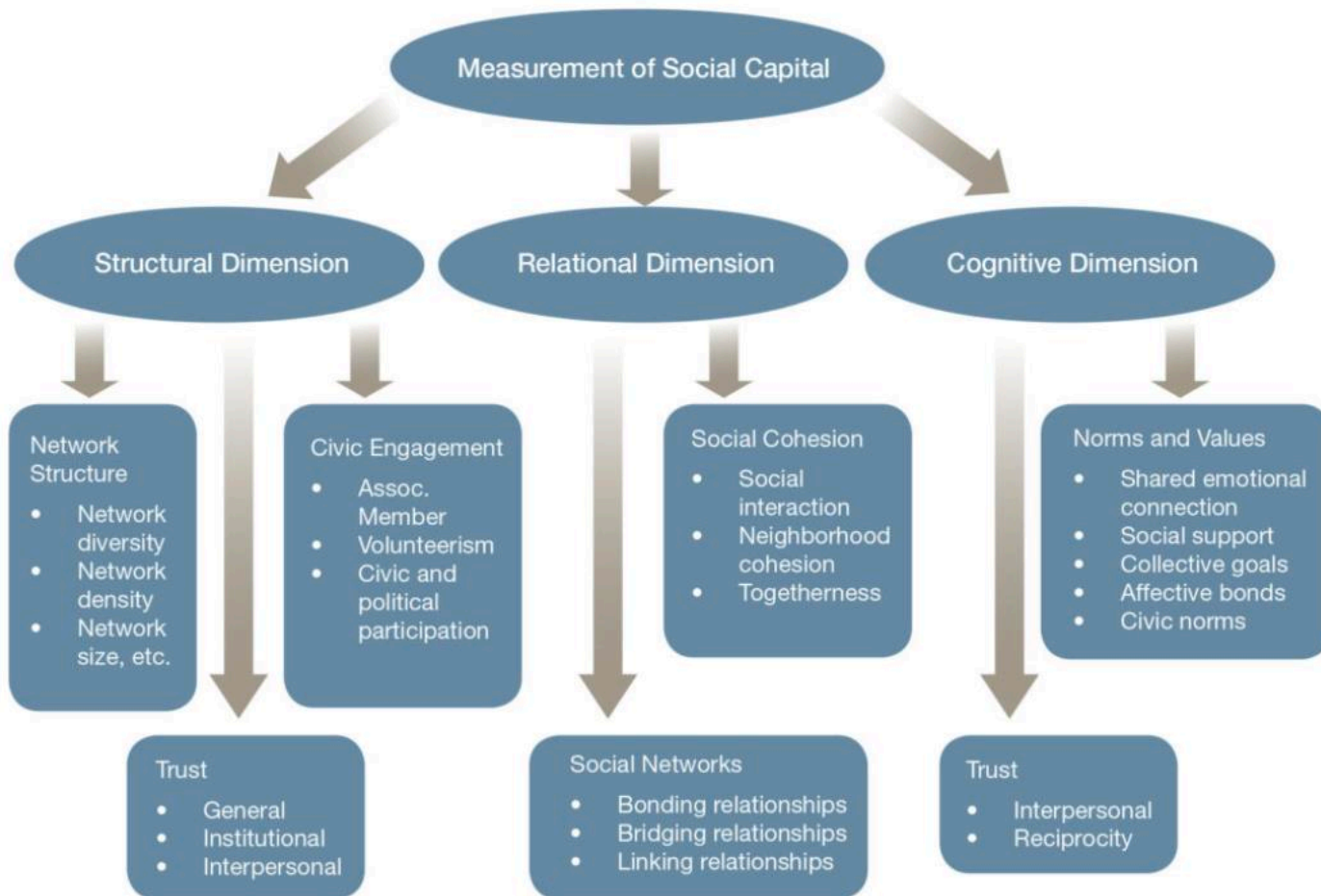


# Social Capital

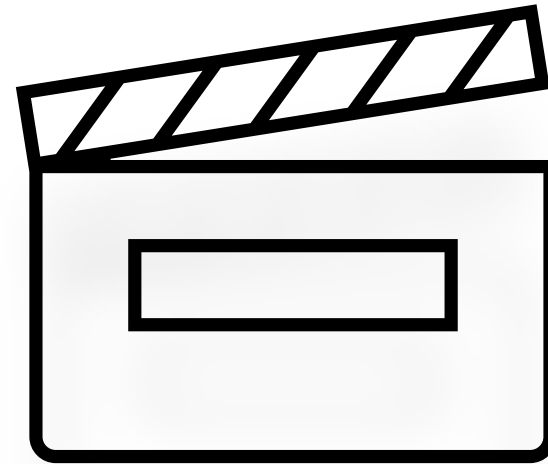
| Relationship level  | Methods of interaction                                                          | Social capital associated with the interactions |
|---------------------|---------------------------------------------------------------------------------|-------------------------------------------------|
| Private, individual | Personal conversations with close friends, face-to-face/one-to-one interactions | Deep, based on reciprocity                      |
| Public, collective  | conversations and group discussions, multiple interactions                      | Extended, propagation-based (diffusion)         |

## 3 dimensions of social capital (Claridge, 2018; Lee, 2017)

| Dimension  | Definition                                                                                                                                                                                                        | Examples                                                                                                                                               |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| STRUCTURAL | Impersonal configuration of linkages between people or units - relates to the properties of the social system and of the network of relations as a whole                                                          | density, connectivity, hierarchy and appropriability of the network of relationships in any given context, such as a group, organisation, or community |
| RELATIONAL | The nature of social relationships developed through interaction between members - relates to the characteristics and qualities of personal relationships such as trust, obligations, respect and even friendship | trust and trustworthiness, norms and sanctions, obligations and expectations, and identity and identification                                          |
| COGNITIVE  | Shared systems of values, beliefs, perspectives, representations and interpretations - elates resources providing shared representations, interpretations, and systems of meaning among parties                   | Language, codes, shared goals, culture, vision, narratives, expertise                                                                                  |



**SURSA: Claridge(2017),**  
<https://www.socialcapitalresearch.com/measure-social-capital/>



## Other perspectives on understanding social capital...

1. Reza Varzidehka, 2023- What is social capital and how can you build yours? (TEDxKerrisdale ) <https://youtu.be/wDshG3HYbul?si=dO2Bxz-jhO4G7Xvn> (accessed 18.10. 2023)
2. Al Condeluci, 2014- Social capital and the power of relationships (TEDxGrandviewAve) <https://youtu.be/qaBUV2J0ax4> (accessed 18.10. 2020)

# Perspectives on social capital

- Social capital as the nature and extent of networks and associated norms of reciprocity → social capital, enables individuals to gain access to resources - ideas, information, money, services, benefits - and to have accurate expectations about the behaviour of others by virtue of their participation in relationships that are themselves the product of networks of association (Putman, 2000).
- Burt, Lin → Social capital refers to the resources (e.g., information, social control) that circulate through networks, not the network structure itself.
- Putnam sees social capital as the "wires" (or **social infrastructure**), while Burt and Lin see it as the "electricity" (or **social resource**)



Access to power  
Social inequalities



# Bonding

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- **Bonding** social networks → **Bonding social capital** : manifests itself through trust and relationships placed at the level of the primary group or groups of belonging - it ensures the cohesion of these groups and contributes to the creation of a familiar environment in which access to goods and information is guaranteed. (Voicu, 2015, p.60)
- strong ties - e.g. between members of the same family or within an ethnic group; this type of social capital is about the day-to-day life of everyone and is indispensable for living (*getting by*);
- „to trusting and co-operative relations between members of a network who see themselves as being similar, in terms of their shared social identity” (Szreter și Woolcock, 2004, p.654)





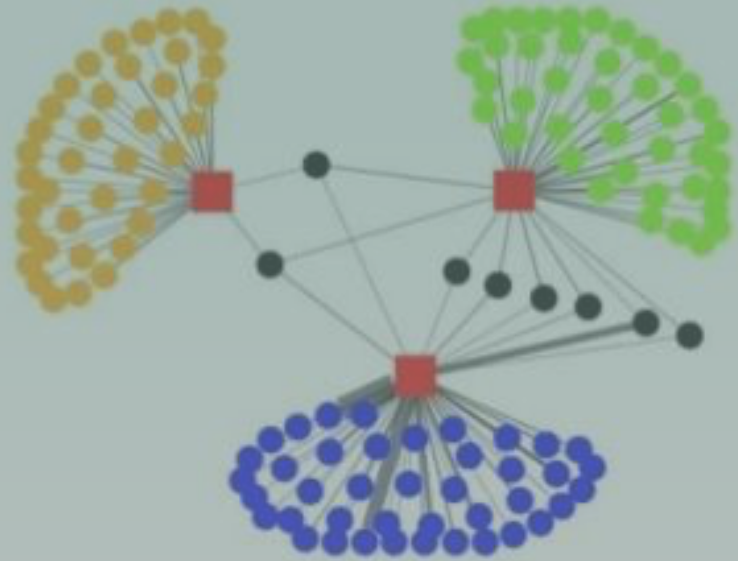
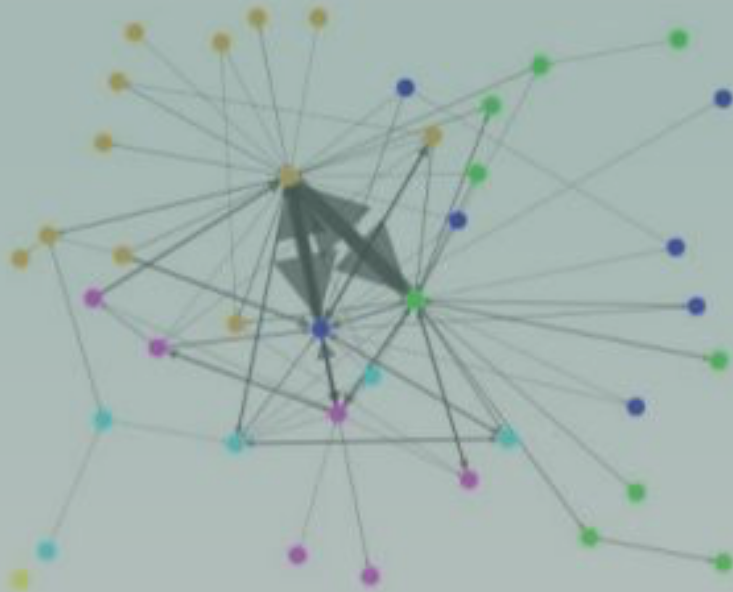
# Bridging

- **Bridging** social networks → **bridging social capital**: is that which manifests itself through participation in associations, through frequent meetings with friends, through generalized trust (in all people, not only in specific groups) and through trust in institutions; it unites through trust and relationships individuals from different groups, allowing them to develop common actions, to access things and to learn information under the control of people much different from those in the group of primary belonging (e.g., family) - (Voicu, 2015, pp.60-61)
  - weaker, less dense but overlapping and interlocking ties - e.g. business partners, circle of acquaintances, friends, friends of friends, generally contacts between people belonging to less close groups; it is this type of social capital that favours progress (getting ahead), by putting into motion social energies in different areas and groups
  - comprises relations of respect and mutuality between people who know that they are not alike in some socio-demographic (or social identity) sense (differing by age, ethnic group, class, etc)- Szreter și Woolcock, 2004, p. 655

1. Members of a network: interconnected individuals with strong ties to each other – **BONDING SOCIAL CAPITAL**

3. Those who interact and form links between groups – **BRIDGING SOCIAL CAPITAL**

2. Groups in a community („small churches”). Weak ties - few group members interact with each other

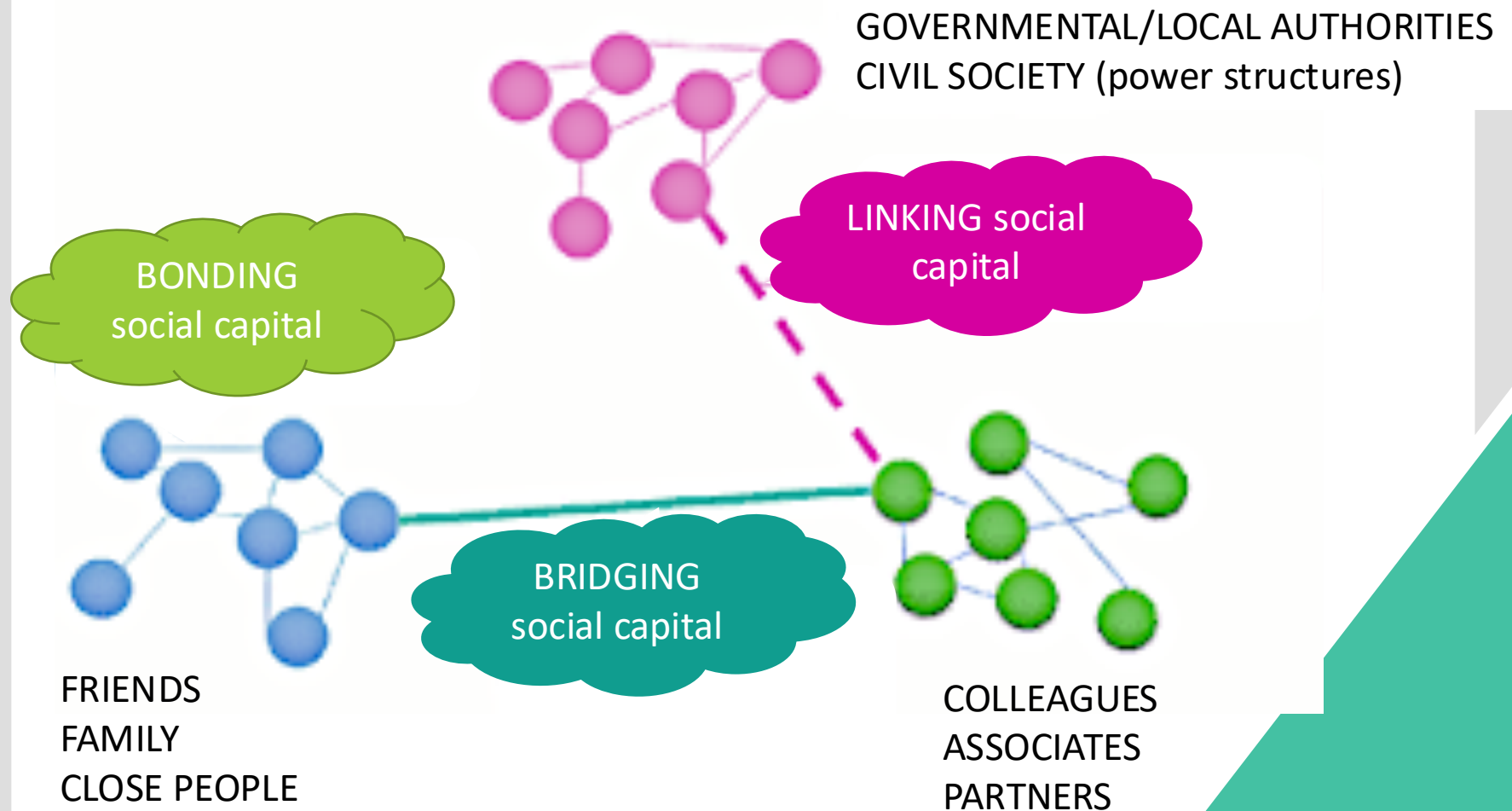


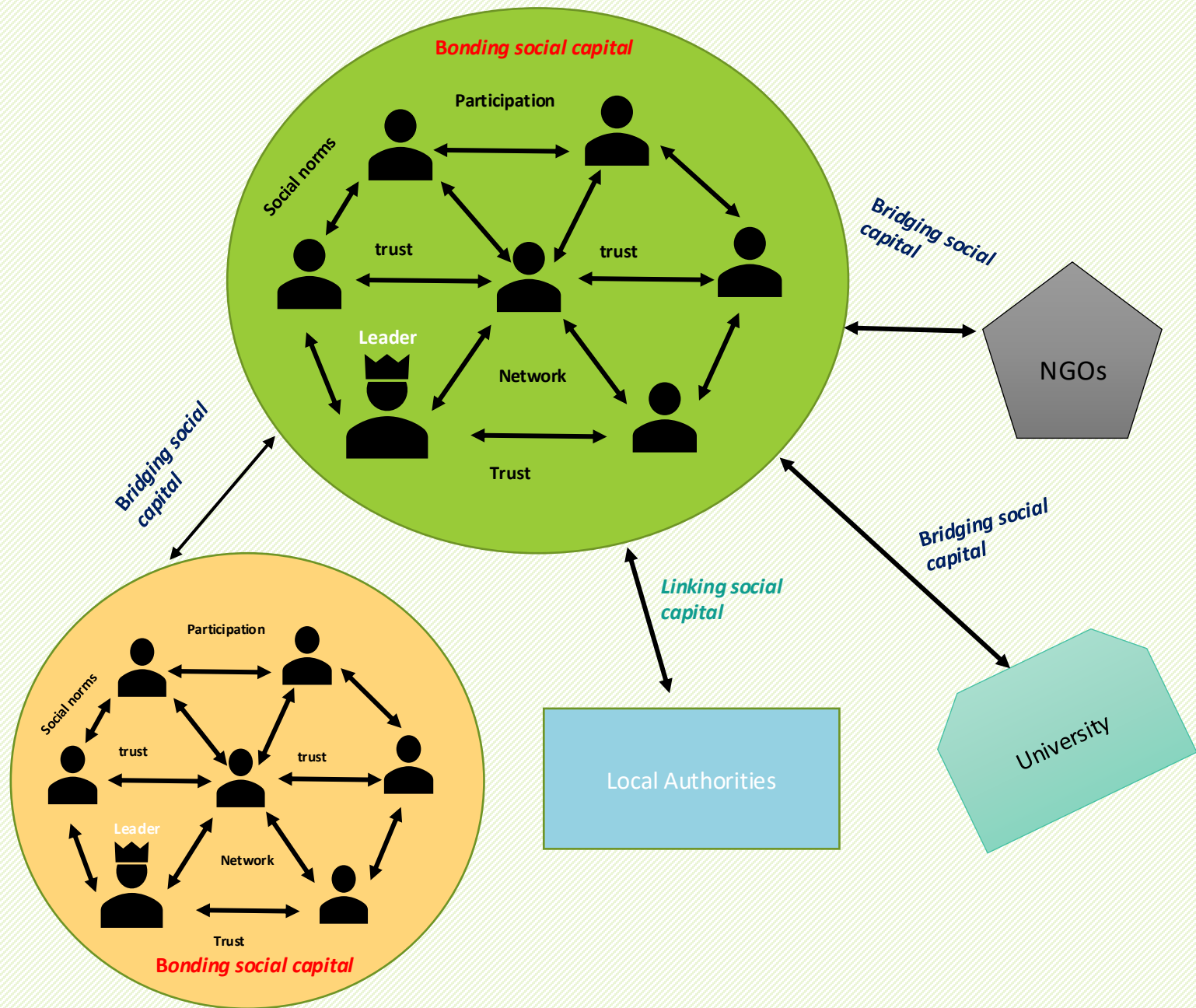
# Linking

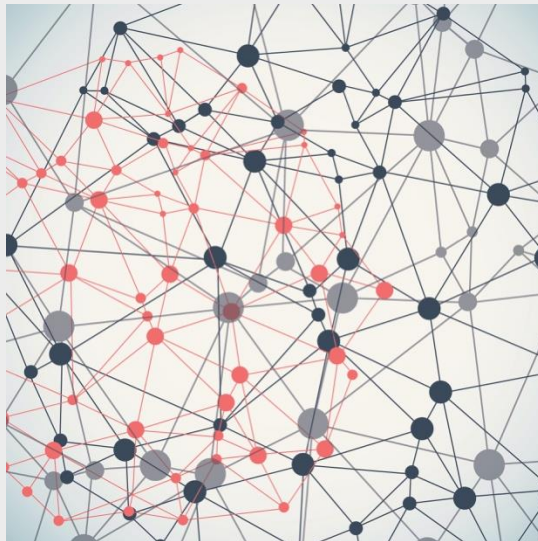
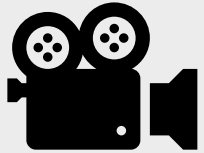
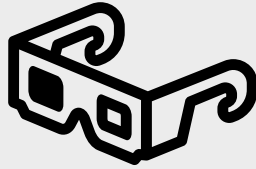
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- **Linking social networks → Linking social capital:** refers to connections between individuals and groups of individuals at different levels of influence and social status; for example, connections between the political elite and the general public or between individuals belonging to different social classes
  - the ability to draw on and manage resources, ideas and information from institutions beyond the community - is mainly the realm of formal institutions
  - „ norms of respect and networks of trusting relationships between people who are interacting across explicit, formal or institutionalized power or authority gradients in society. ” (Szreter & Woolcock, 2004, p. 655)









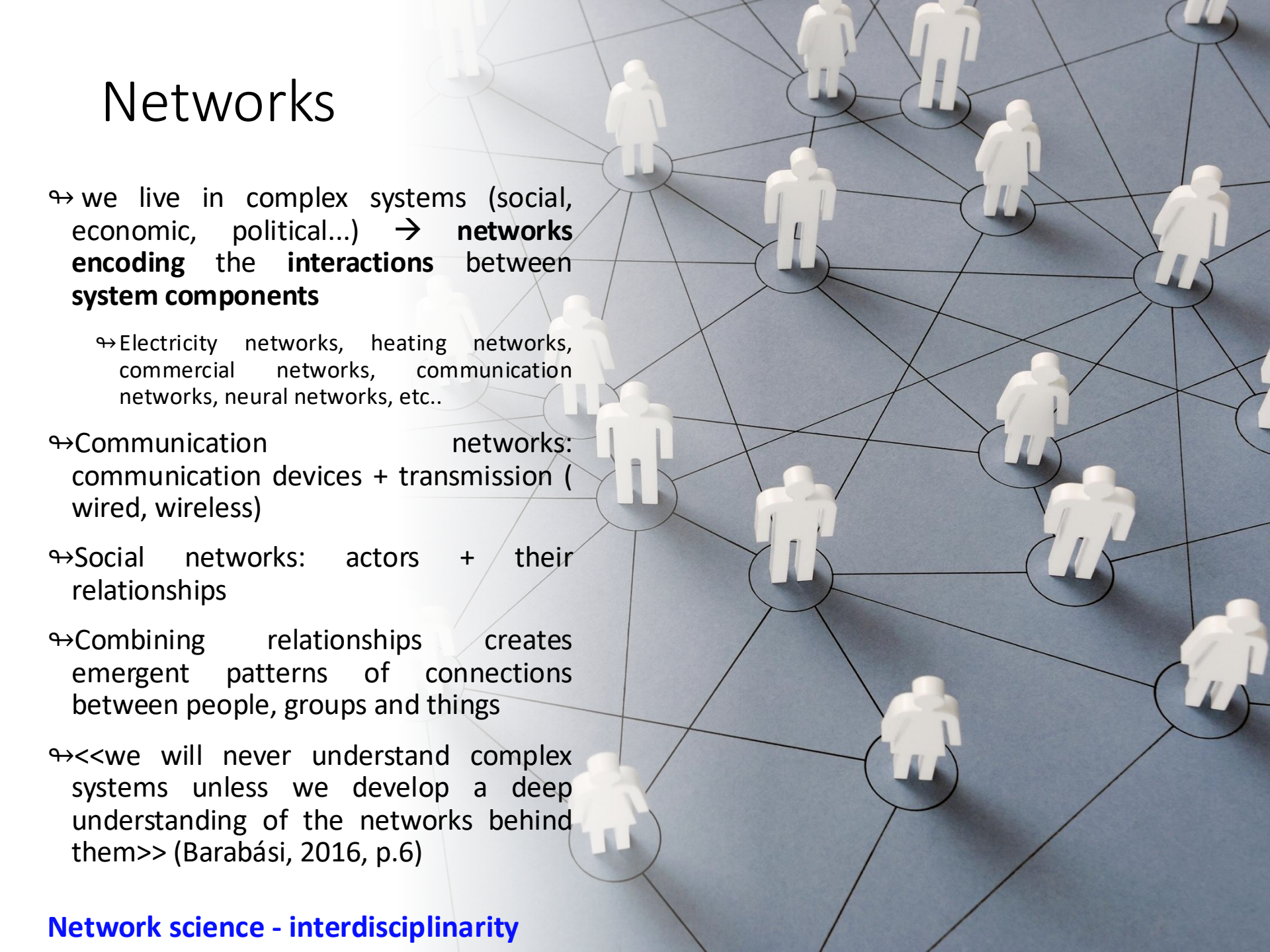
Manuel Lima (2012):  
The Power of  
Networks (RSA  
ANIMATE)

<https://youtu.be/nJmGrNdJ5Gw>

# Some research based on the study of types of social capital

- Han, S., Grace Oh, E., & “Pil” Kang, S. (2022). Social Capital Leveraging Knowledge-Sharing Ties and Learning Performance in Higher Education: Evidence From Social Network Analysis in an Engineering Classroom. *AERA Open*, 8. <https://doi.org/10.1177/23328584221086665>
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# Networks

A network diagram with white human figures as nodes connected by black lines on a blue background. The figures are arranged in a circular pattern, with lines connecting them to form a complex web of relationships. The background is a solid blue color, and the figures are white with black outlines.

↔ we live in complex systems (social, economic, political...) → **networks encoding the interactions between system components**

↔ Electricity networks, heating networks, commercial networks, communication networks, neural networks, etc..

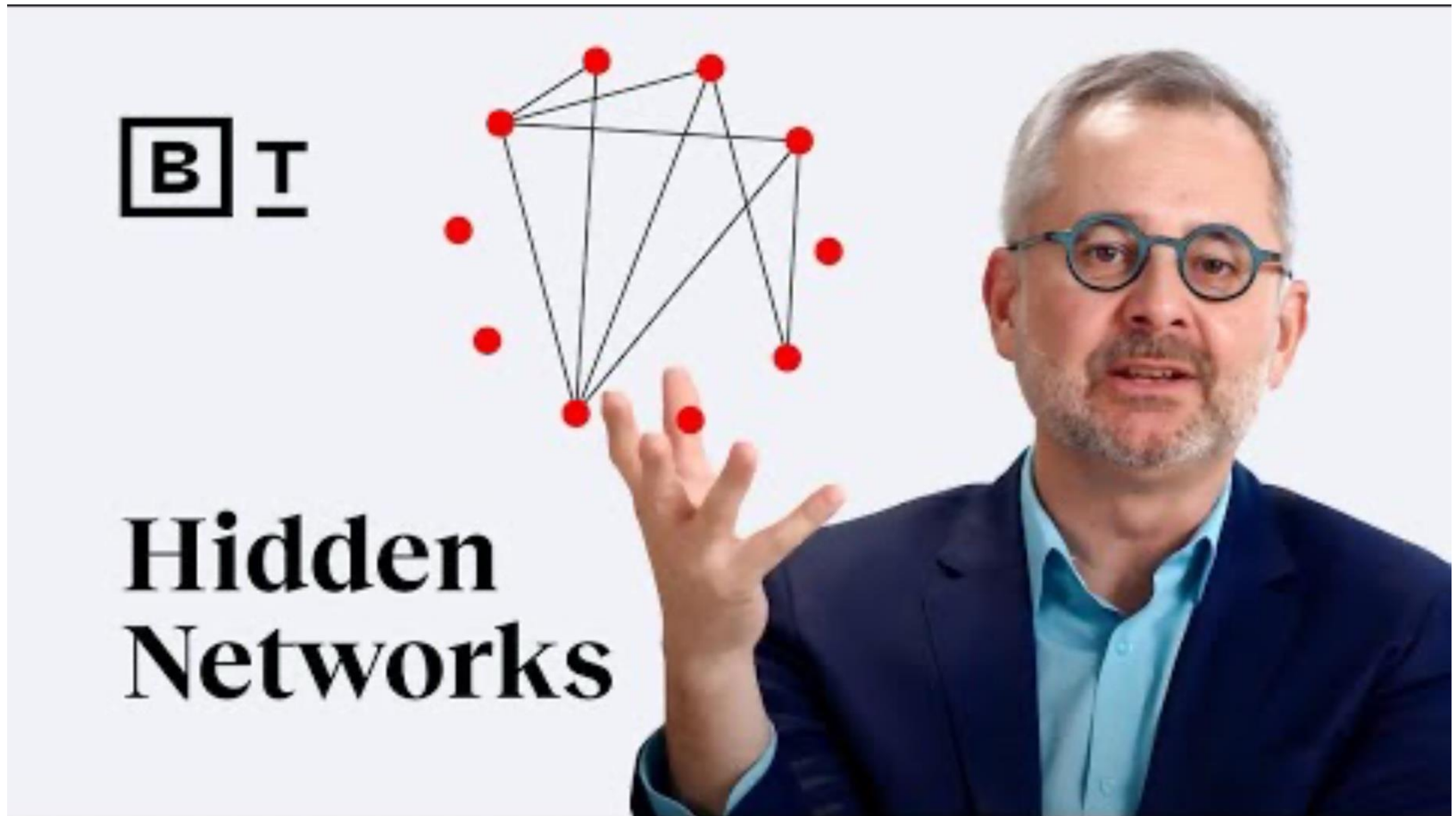
↔ Communication networks: communication devices + transmission (wired, wireless)

↔ Social networks: actors + their relationships

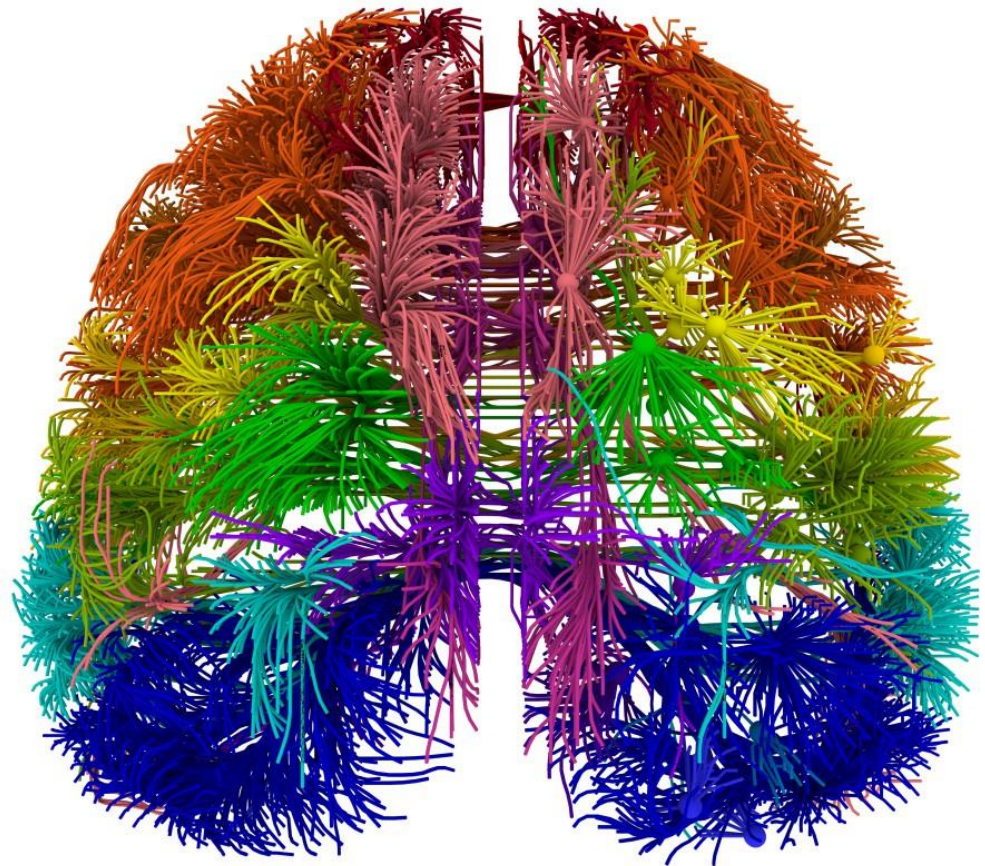
↔ Combining relationships creates emergent patterns of connections between people, groups and things

↔ <<we will never understand complex systems unless we develop a deep understanding of the networks behind them>> (Barabási, 2016, p.6)

<https://youtu.be/RfgjHoVCZwU?si=DyXaaJqFa q9W2H8c>

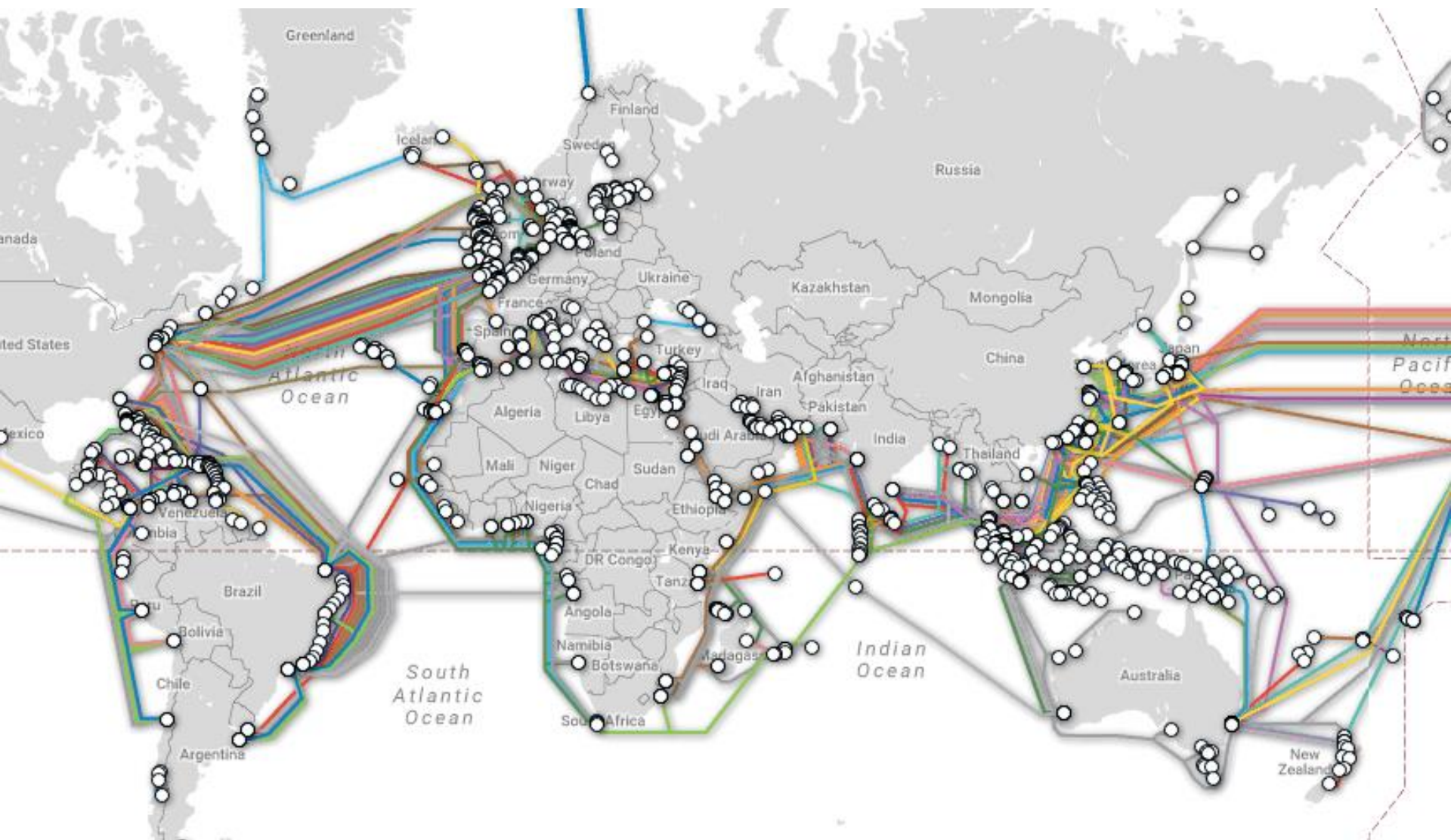


Source: Allen Institute for Brain Science(2014): First 3D diagram of a mouse brain;  
Oh, S., Harris, J., Ng, L. et al. A mesoscale connectome of the mouse brain. *Nature* 508, 207–214 (2014).  
<https://doi.org/10.1038/nature13186>



The Human Connectome Project: <http://www.humanconnectomeproject.org>





Source: Telegeography (2021): Global Internet backbone map

See: <https://www.submarinecablemap.com>

# Ties in the network

Network= a collection of dots joined by lines(Newman, 2014).



| Network     | Node                                 | Tie                                  |
|-------------|--------------------------------------|--------------------------------------|
| Electricity | Generating stations or substations – | Transmission lines (cables, pipes..) |
| Neuronal    | Neurons                              | Synapse                              |
| Internet    | Computer or router –                 | Wired or wireless connection         |
| www:        | Web pages –                          | Hyperlink                            |
| Facebook    | Profile/user –                       | „Friendship”                         |

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